

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/FINAL/B_DEB_FLYe")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 = 8484)
seeds <- c(C1 = 3313, C2 = 6666, C3 = 1212)

text_priors <- '
Distrib (Fm,          TruncNormal,          0.1,      0.2, 0.0001,      0.8);
Distrib (pAm,         TruncNormal_cv,        100,      0.2,      40,      2000);
Distrib (r_pAm_pM,    TruncNormal_cv,        0.8,      0.1,      0.1,      2);
Distrib (v,           TruncNormal_cv,        0.18505,   0.1,      0.001,      2);
Distrib (kap,         Uniform,                0,          1);
Distrib (Ehp,         TruncNormal_cv,        100,      0.2,      20,      300);
Distrib (E_coc,       TruncNormal_cv,        30,      0.2,      1,      300);
Distrib (rOM_ClXHorse, LogUniform,          0.00001,   0.3);
Distrib (dv,          TruncNormal_cv,        2,      0.5,      0,      10);
Distrib (de,          LogUniform,          0.000000001, 0.1);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihoood <- 'Likelihood (Weight,          Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE
RTOL <- 1e-7
ATOL <- 1e-6

# makemcsim DEBcali.model

# f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihoood, NbIter)

# ./mcsim.DEBcali DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

	Fm.1.	pAm.1.	r_pAm_pM.1.	v.1.	kap.1.	Ehp.1.
Result_mode	0.21625900	298.24500	0.58288800	0.21239100	0.95062700	20.497700
2.5%	0.13552295	258.51800	0.47479225	0.17033080	0.89311900	20.299630

97.5%	0.29406300	316.23400	0.58920070	0.23707100	0.95125900	41.713000
Min	0.00357986	70.83850	0.41499800	0.13785500	0.33632700	20.000100
Max	0.29999900	344.31400	0.88576600	0.27867600	0.95864300	109.028000
Result_mean	0.20895574	286.94577	0.53537083	0.20417282	0.93001412	27.103438
Result_sd	0.04656960	14.92685	0.02891312	0.01708039	0.01581127	5.719617
	E_coc.1.	rOM_ClxFhorse.1.	dv.1.	de.1.	Sigma_W.1.	
Result_mode	1.0090500	0.010175500	8.8000200	1.003740e-09	0.2055970	
2.5%	1.0285720	0.002716491	6.3498265	1.207340e-09	0.1996420	
97.5%	2.2484885	0.013692285	9.3766795	5.743091e-07	0.2718654	
Min	1.0001200	0.000454268	1.6855400	1.003740e-09	0.1714850	
Max	35.4298000	0.023561400	9.9971700	1.139480e-05	5.4031000	
Result_mean	1.4280684	0.008363771	7.8313224	8.372763e-08	0.2334867	
Result_sd	0.4403033	0.002787449	0.7769489	1.892002e-07	0.0514964	

Number of observations, n = 277
 Number of parameters, k = 10
 Log-likelihood, LL = -119.2444
 BIC = 294.729

Potential scale reduction factors:

	Point est.	Upper C.I.
de.1.	1.04	1.05
dv.1.	1.02	1.05
E_coc.1.	1.14	1.42
Ehp.1.	1.13	1.38
Fm.1.	1.00	1.01
kap.1.	1.13	1.38
pAm.1.	1.01	1.02
r_pAm_pM.1.	1.07	1.21
rOM_ClxFhorse.1.	1.01	1.02
Sigma_W.1.	1.01	1.05
v.1.	1.00	1.01

Multivariate psrf

1.08

```

mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {

```

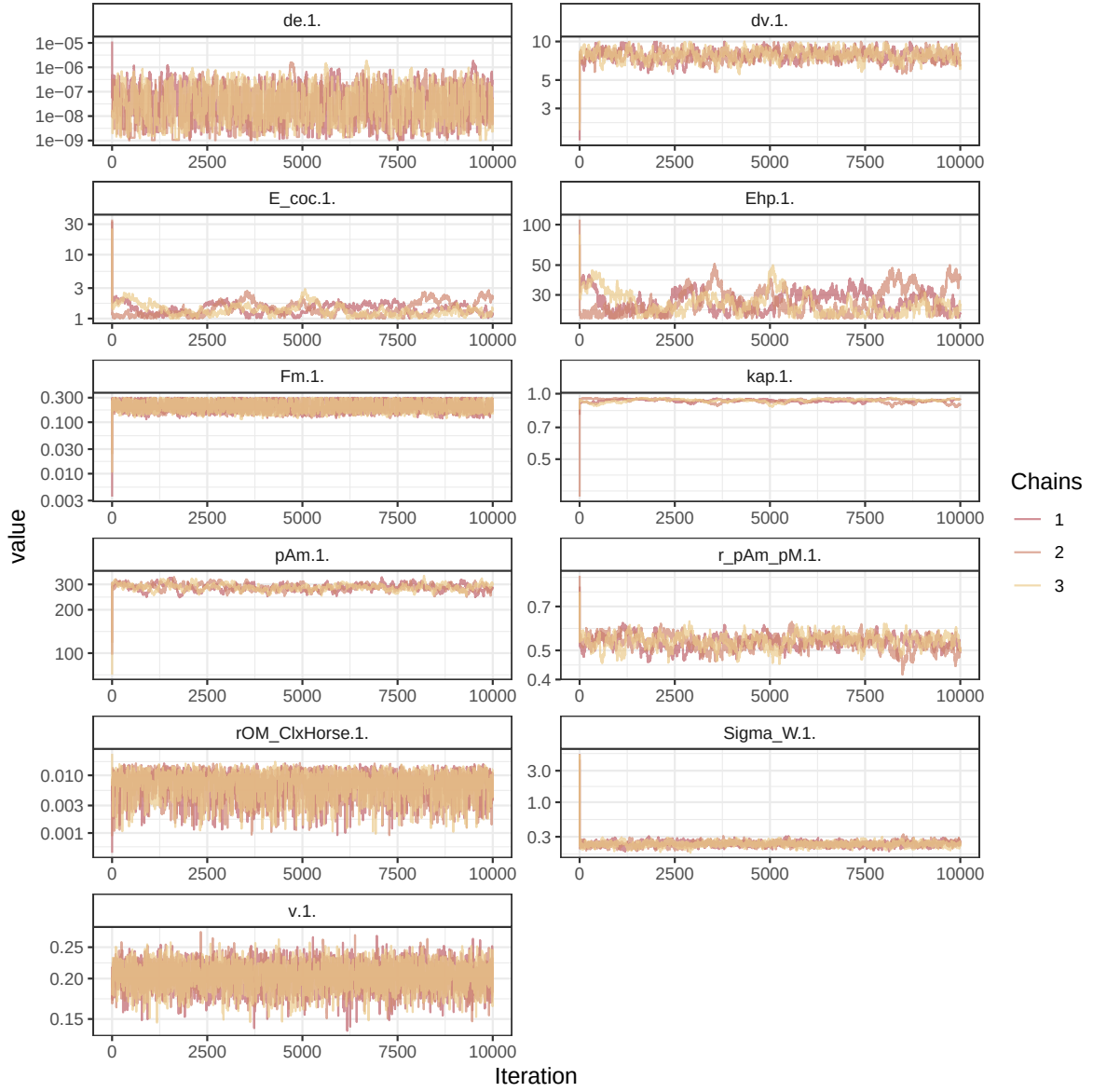


Figure 1: Parameters chains. Only the first iteration and the last `Nb_iter_kept` are represented.

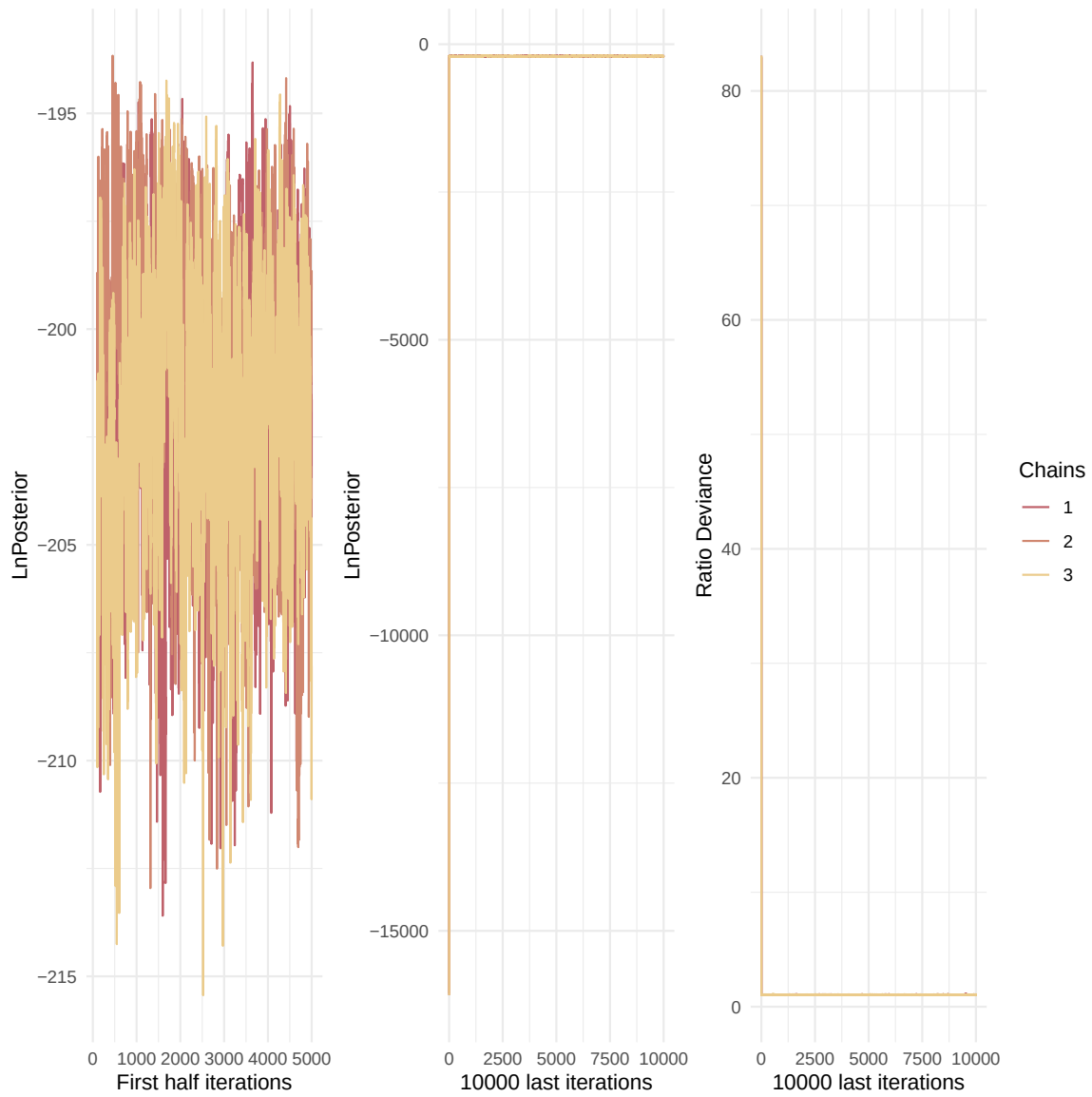


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

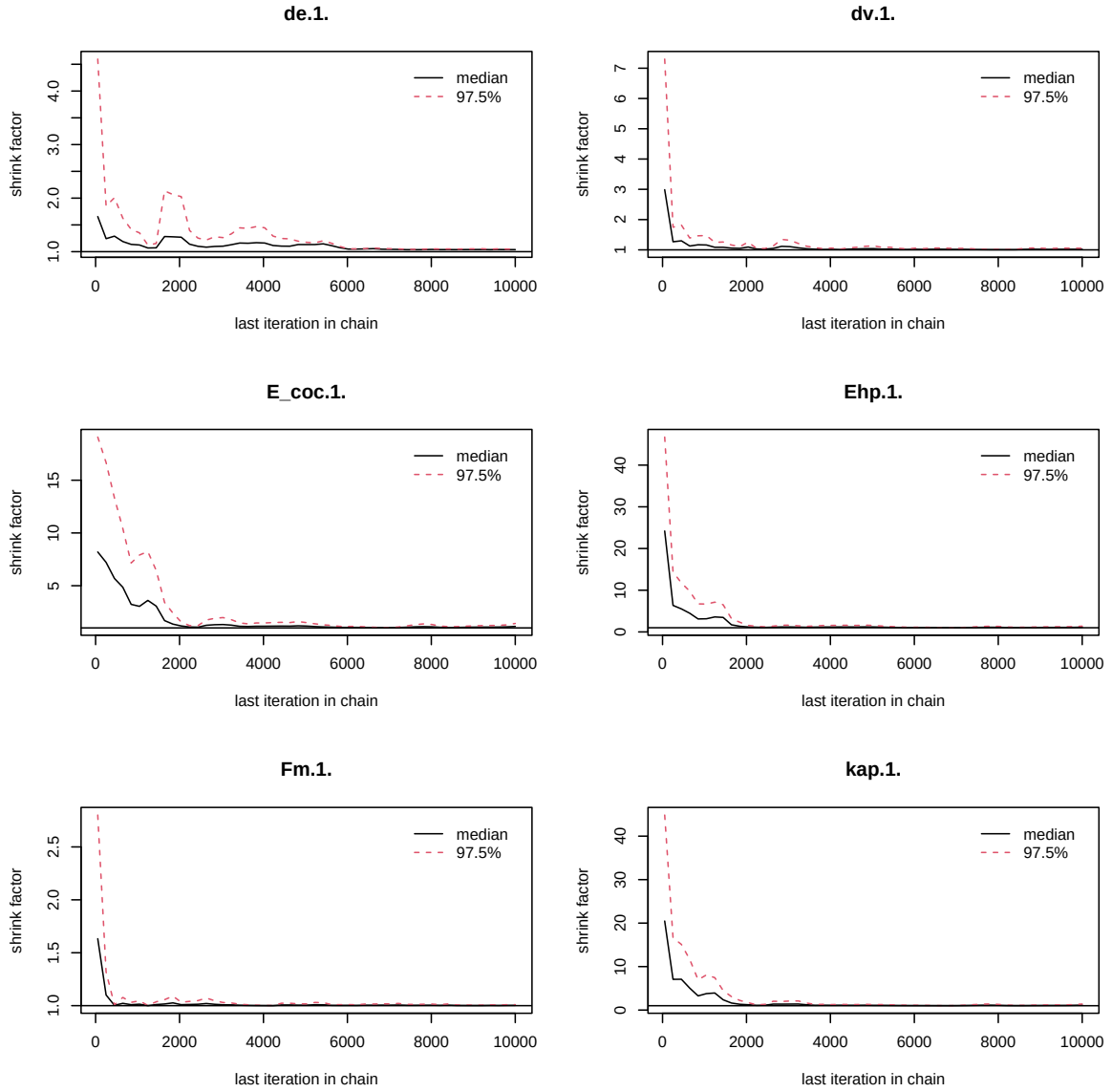


Figure 3: Chains convergence.

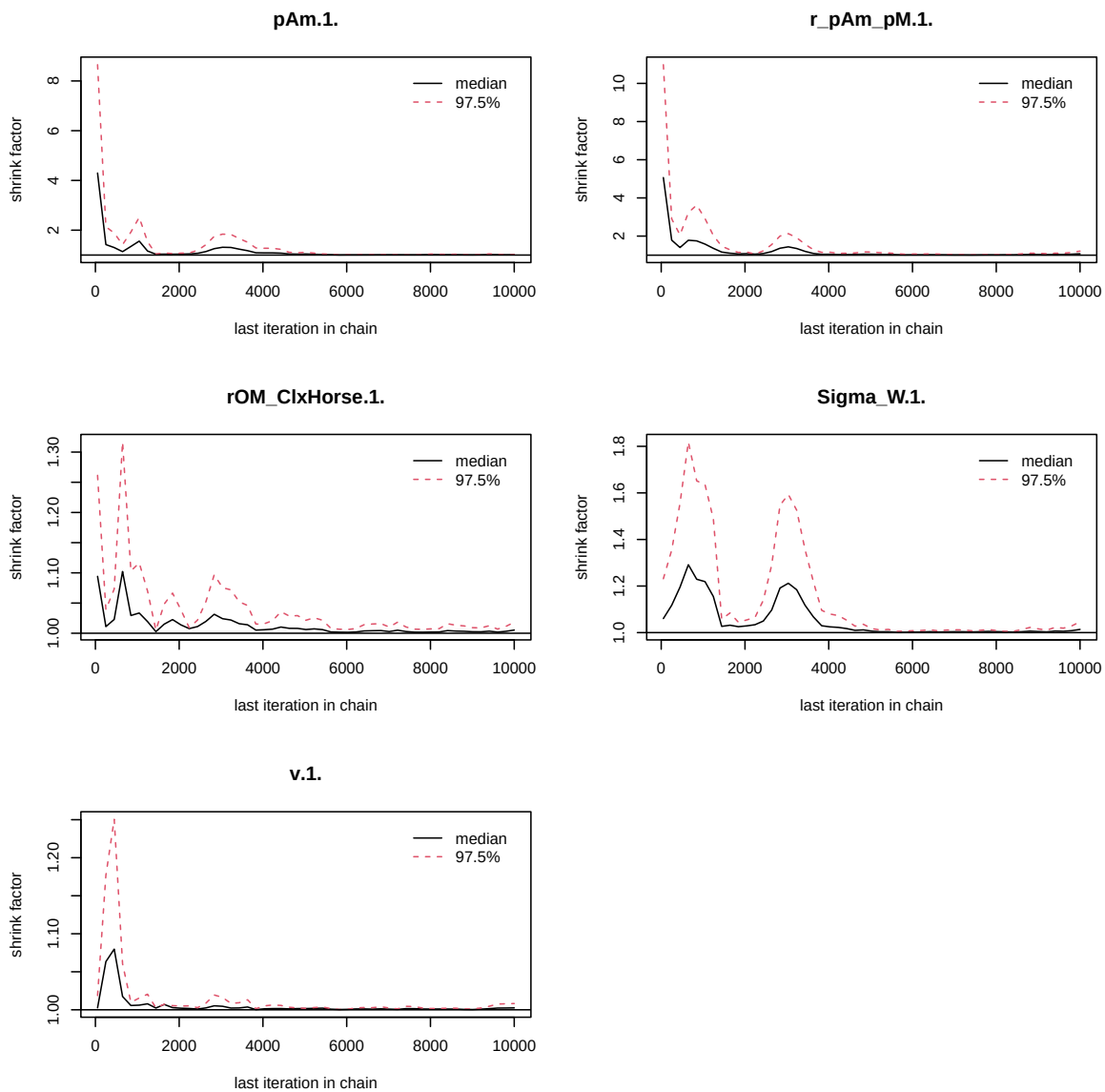


Figure 4: Chains convergence.

```

    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
pairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)

```

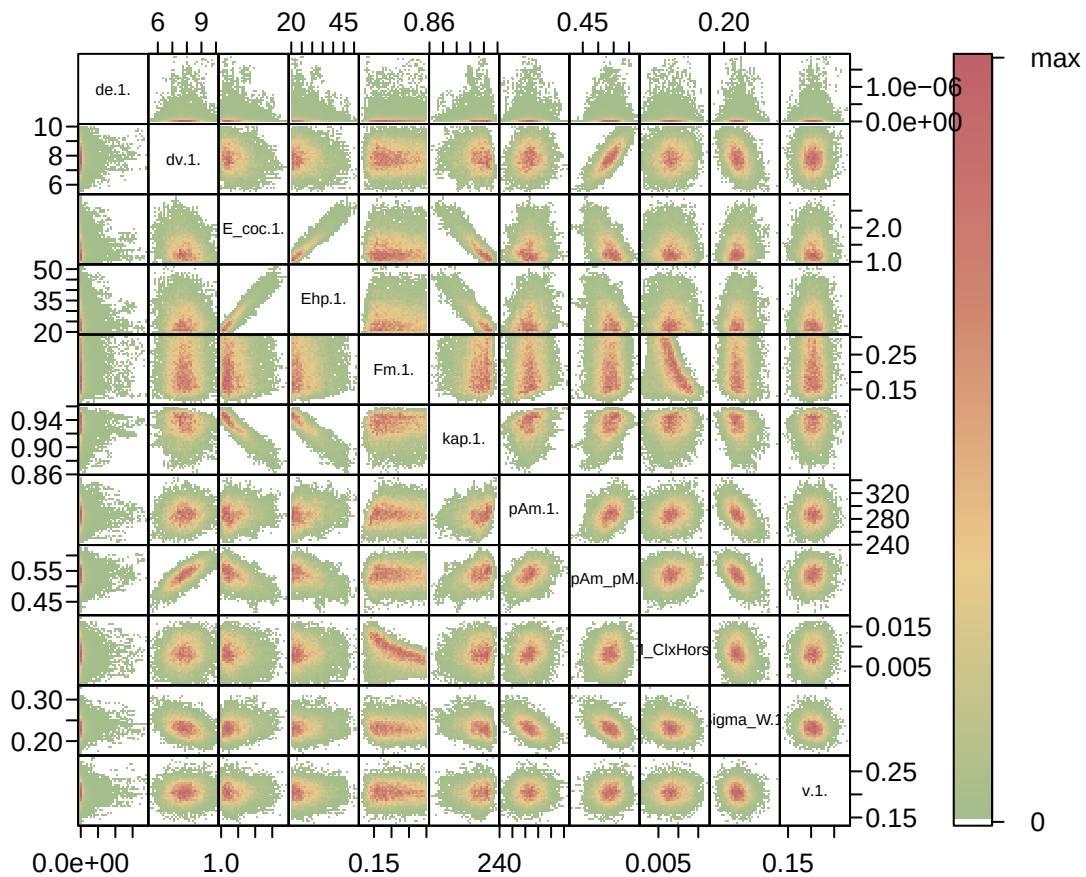


Figure 5: Parameter correlations.

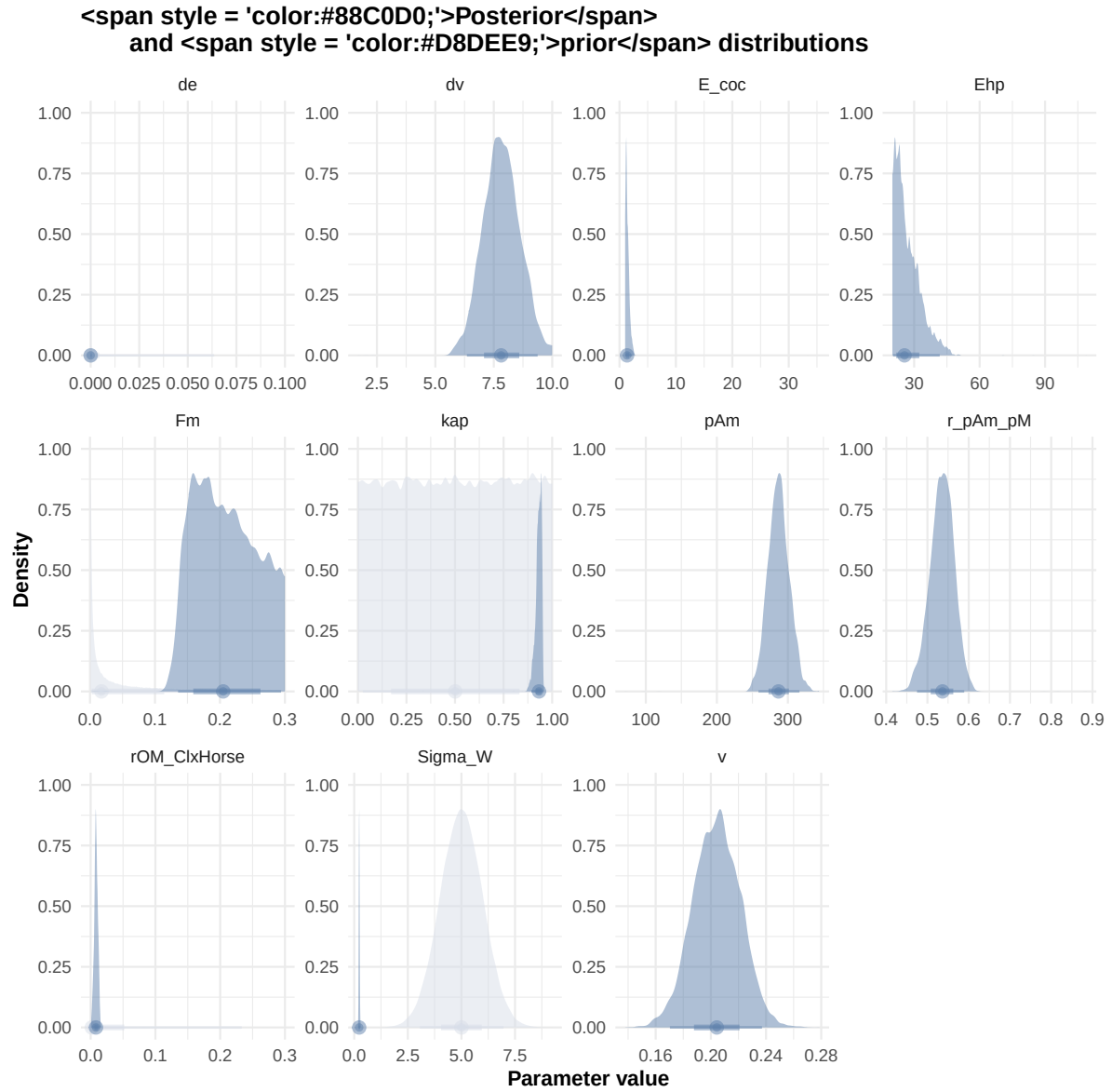


Figure 6: Distributions of the priors and the posteriors.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- TRUE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)

text_param <- Summary_res["Result_mode", , drop = FALSE] %>%
  t() %>%
  as.data.frame() %>%
  tibble::rownames_to_column("param") %>%
  rename(val = 2) %>%
  mutate(
    param = gsub("\\\\.1\\.\\.\\$|\\.1\\.\\.\\.\"", "", param),
    line = glue("{param} = {val};")
  ) %>%
  pull(line) %>%
  paste(collapse = "\n")

Endpoints_print <- "Weight,
  Length_struct,
  Reproduction,
  Energy,
  Organic_matter,
  kapreal,
```

```

    freal,
    Qf_soil,
    Qf_horse,
    X"

step_sim <- 0.01

f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_print)
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_pr

# ./mcsim.DEBcali DEB_sim_full.in

#####
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#####

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times) |>
  mutate(
    Reproduction = case_when(

```

```

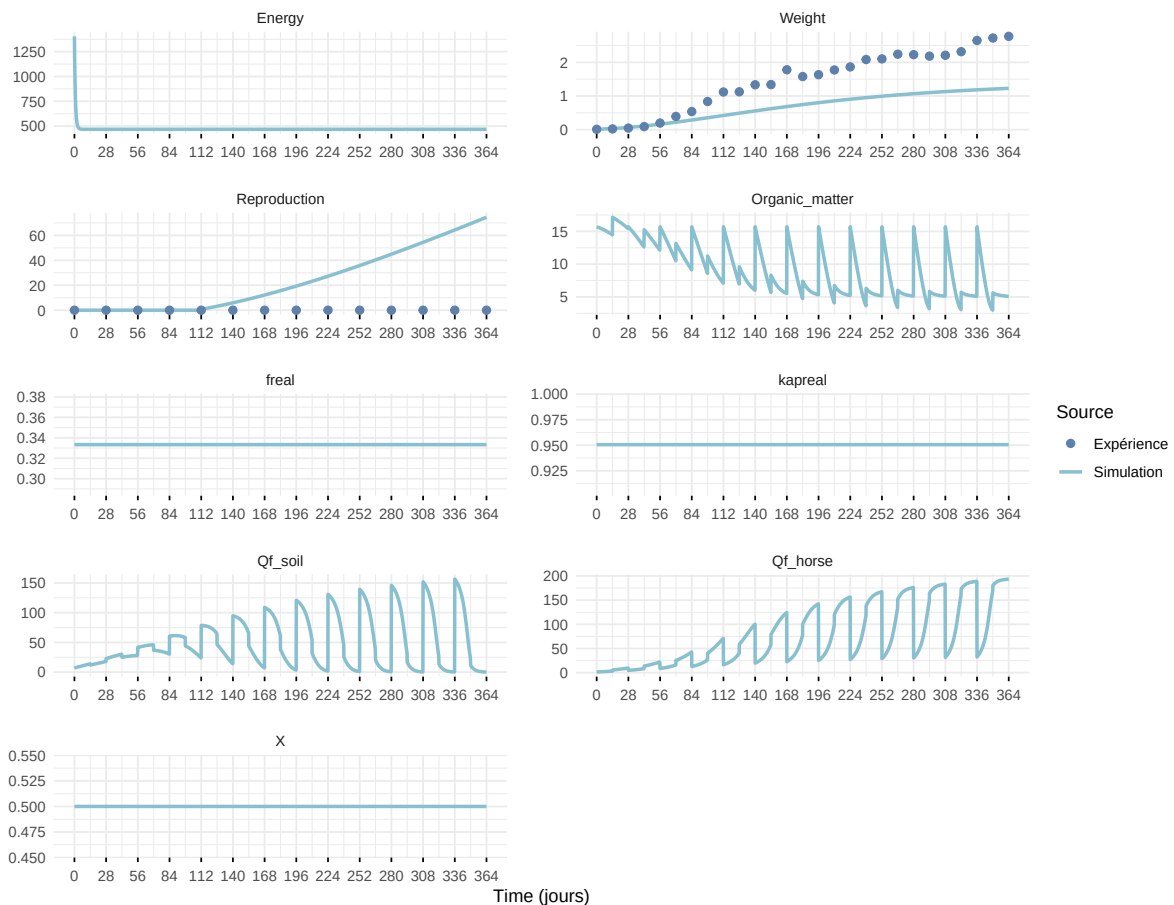
Reproduction <= 5e-6 ~ 0,
.default = Reproduction
)
)

```

Warning: `read\_table2()` was deprecated in readr 2.0.0.
i Please use `read\_table()` instead.

Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpected relationship between 'x' and 'y'.
i Row 1 of 'x' matches multiple rows in 'y'.
i Row 1 of 'y' matches multiple rows in 'x'.
i If a many-to-many relationship is expected, set `relationship = "many-to-many"` to silence this warning.

Expérience : Bart2019_XP1_1



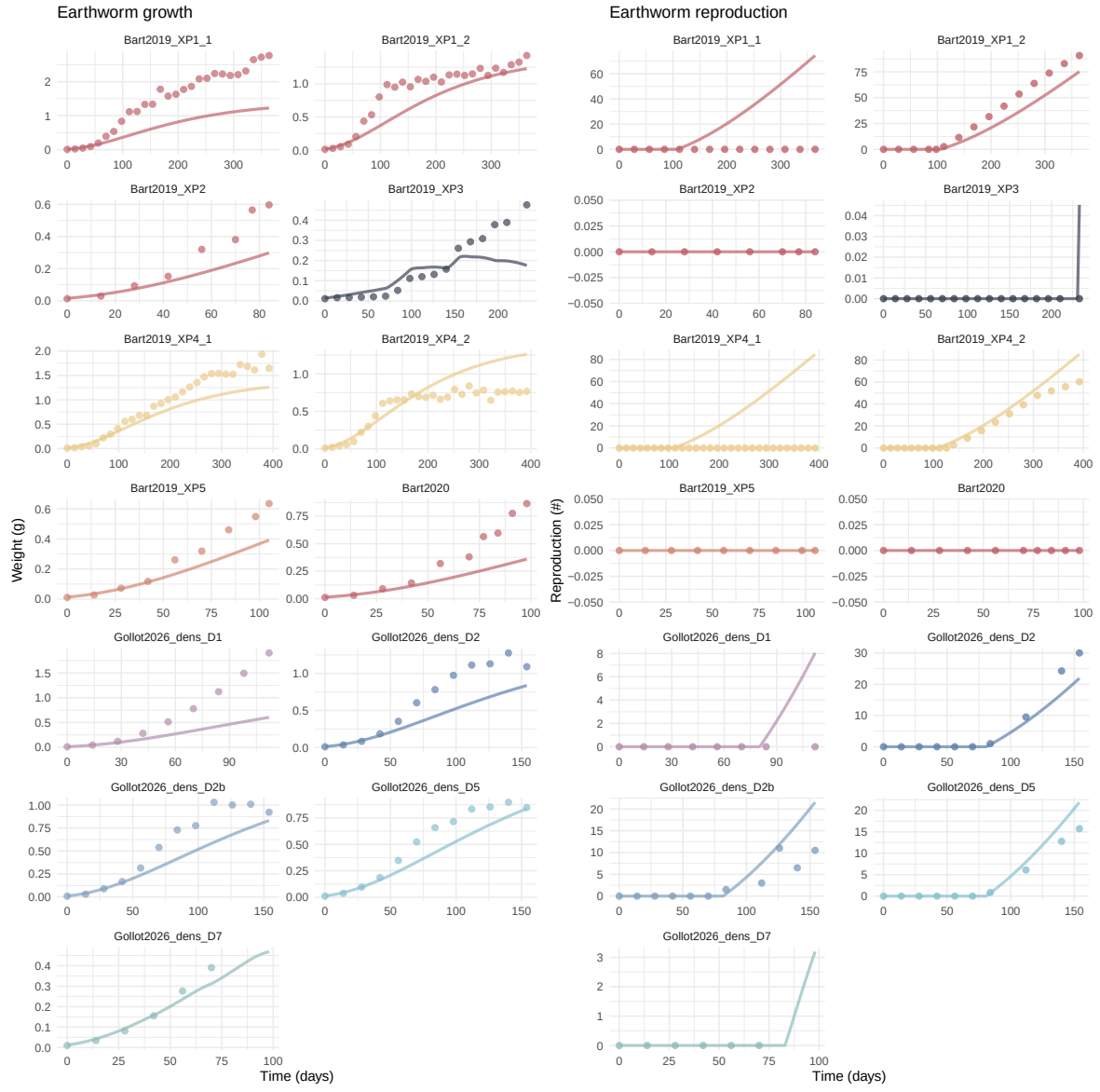
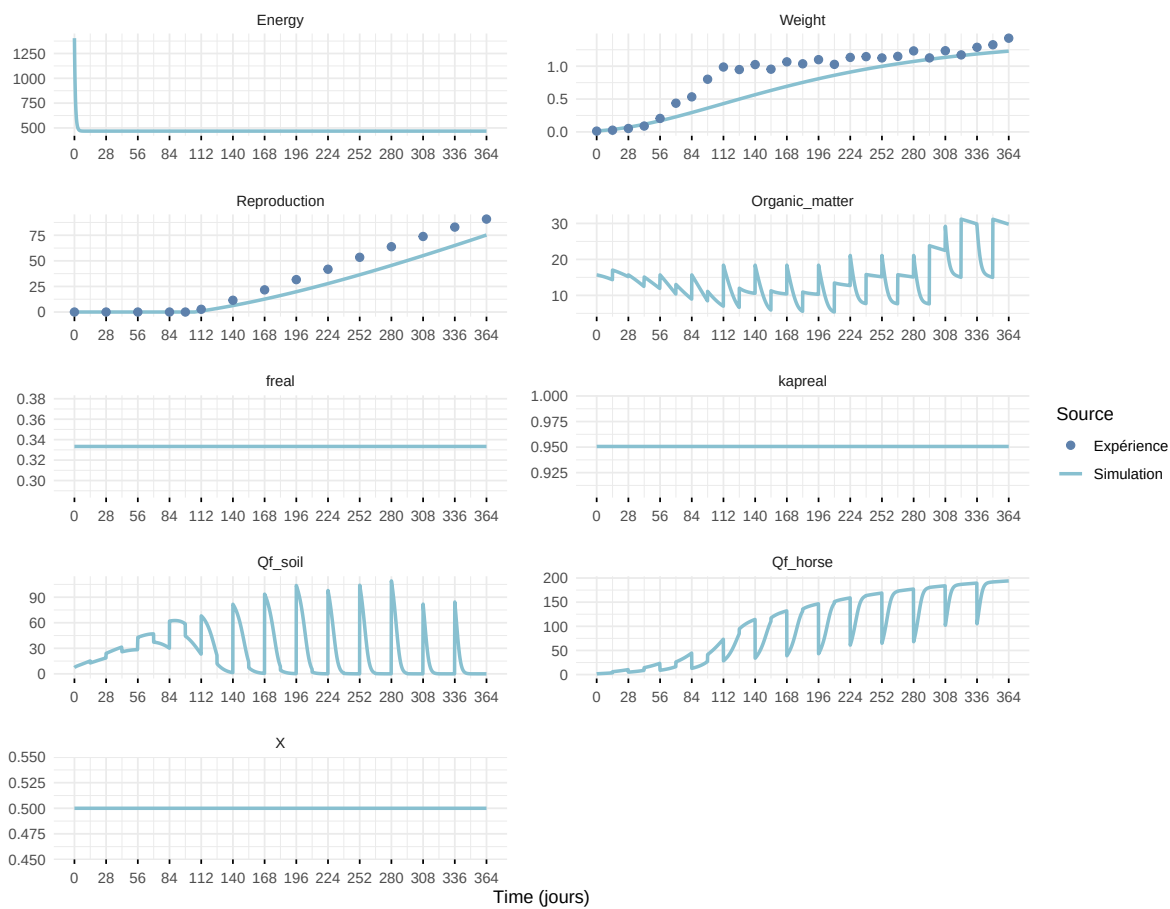
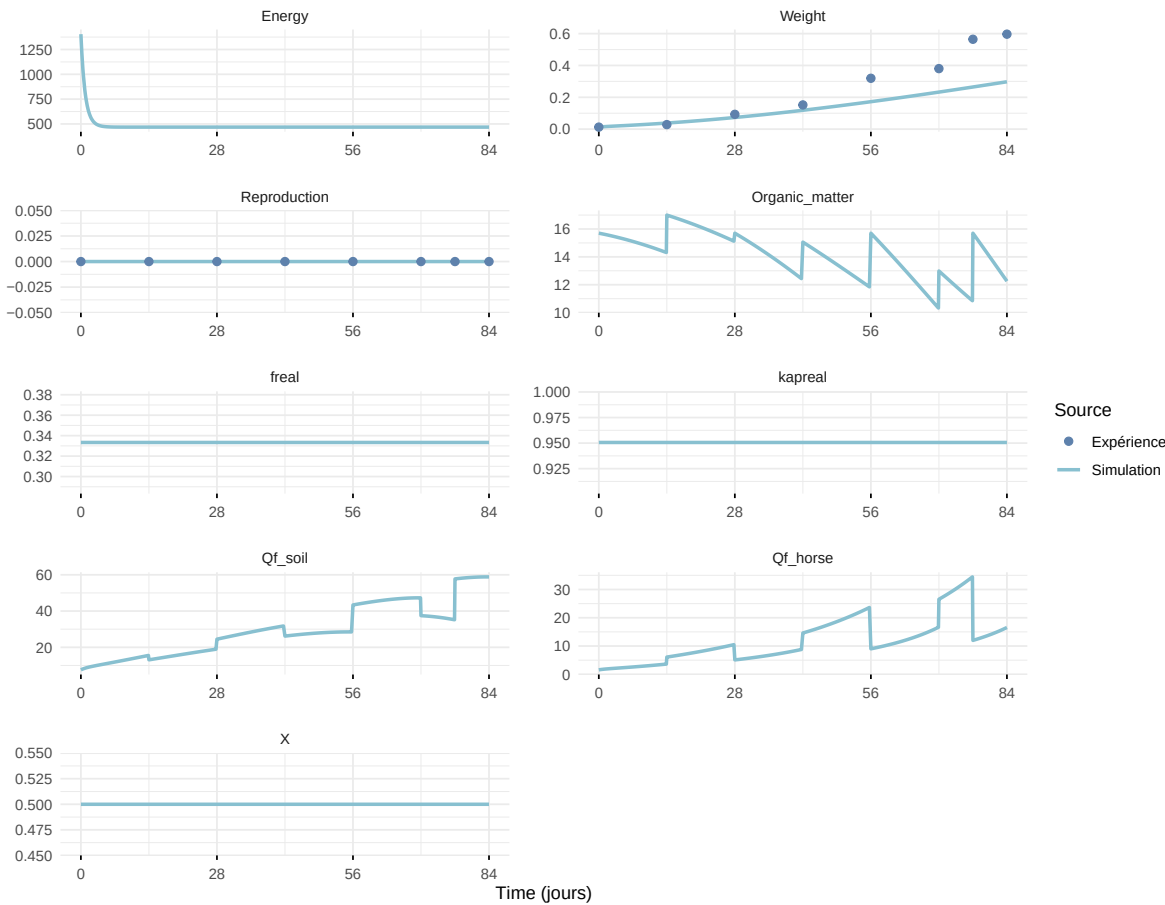


Figure 7: Simulations Weight and Reproduction.

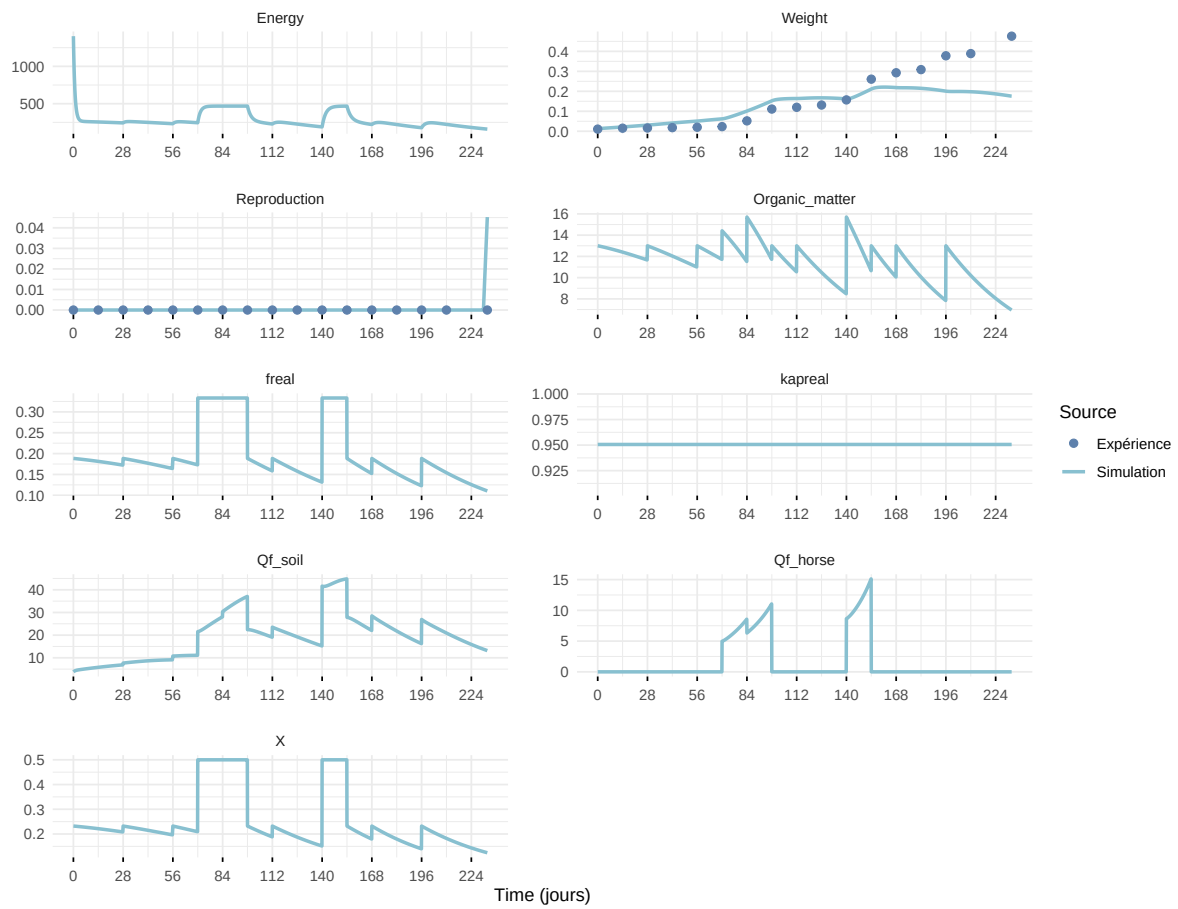
Expérience : Bart2019_XP1_2



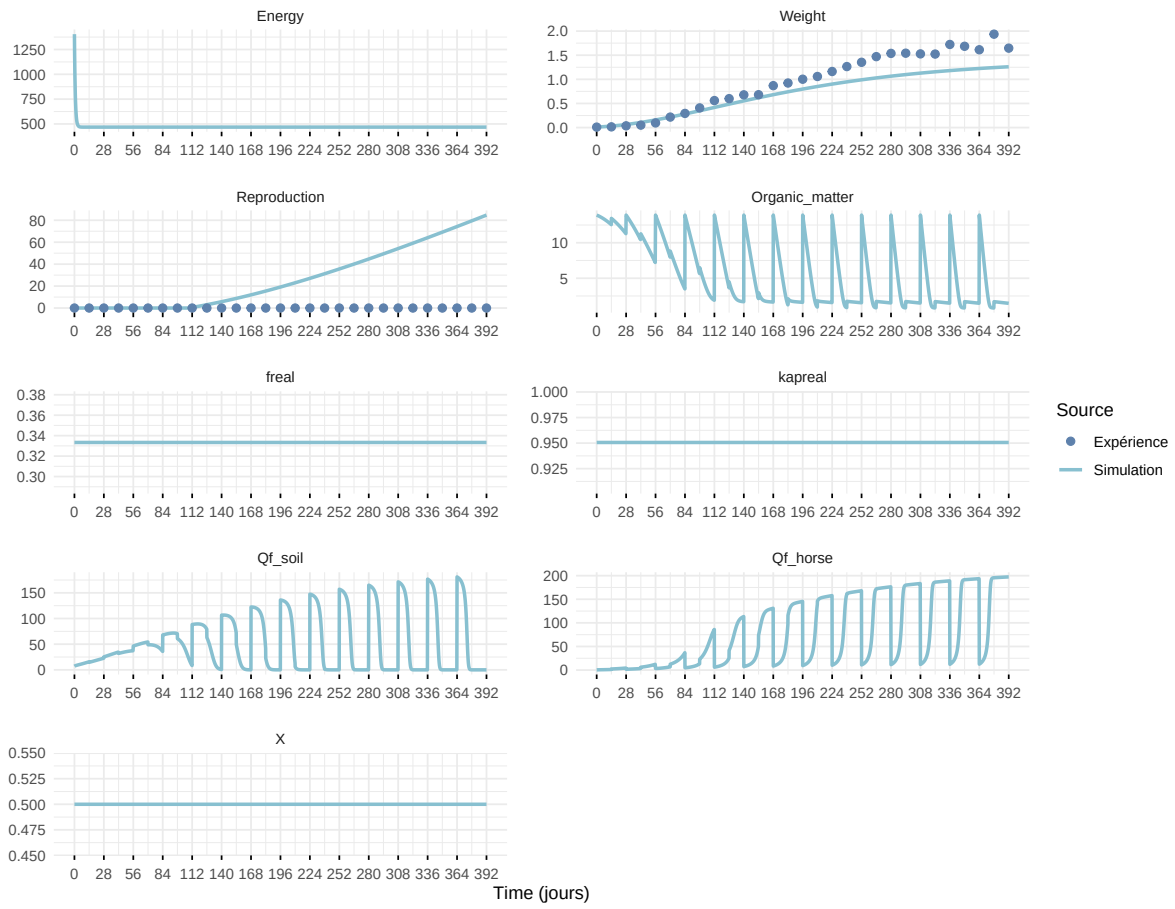
Expérience : Bart2019_XP2



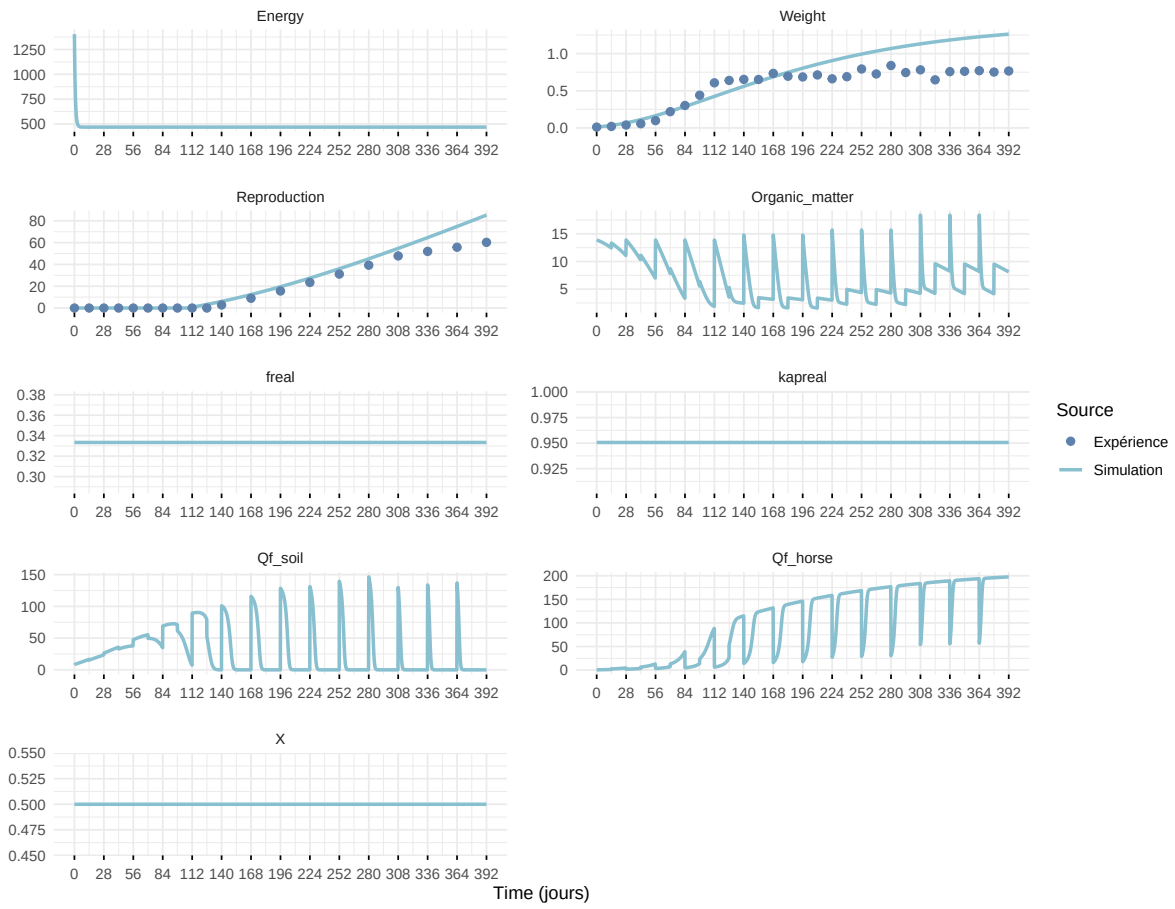
Expérience : Bart2019_XP3



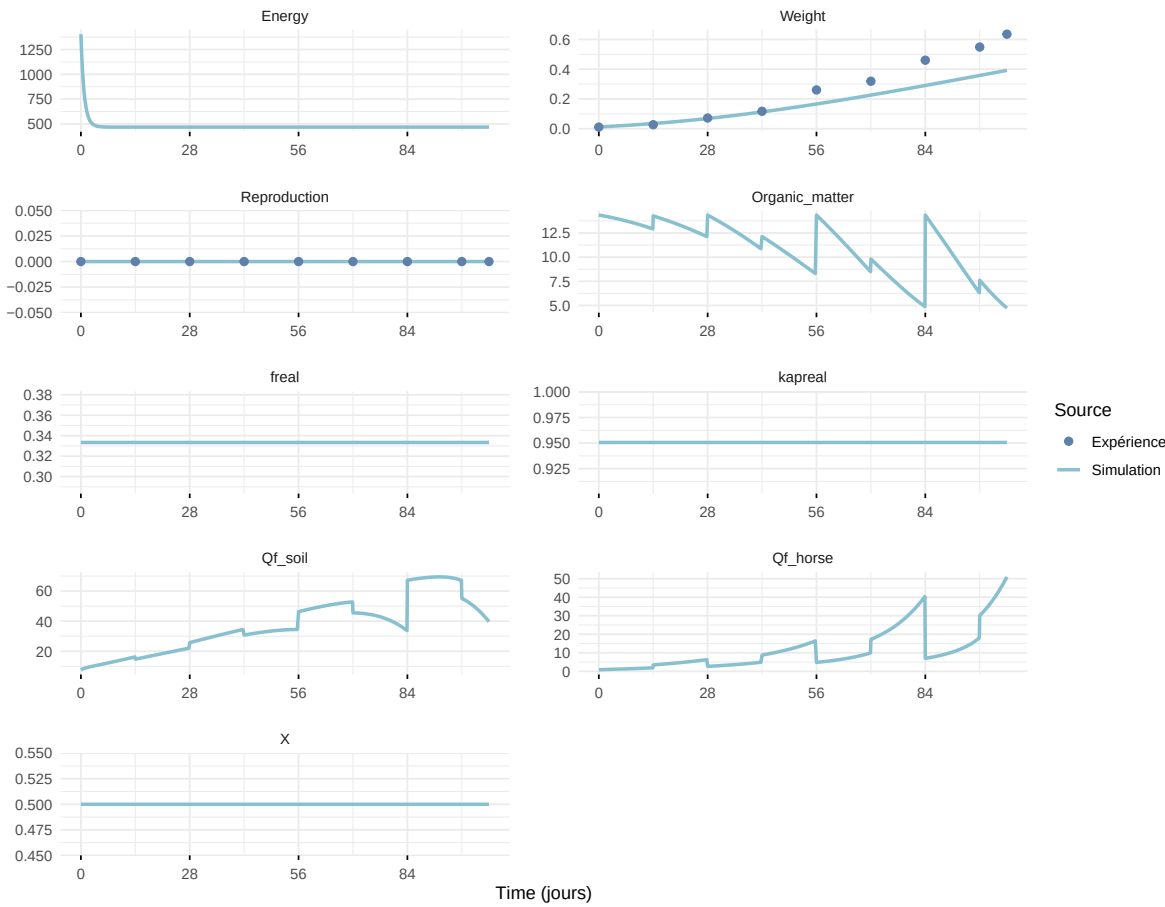
Expérience : Bart2019_XP4_1



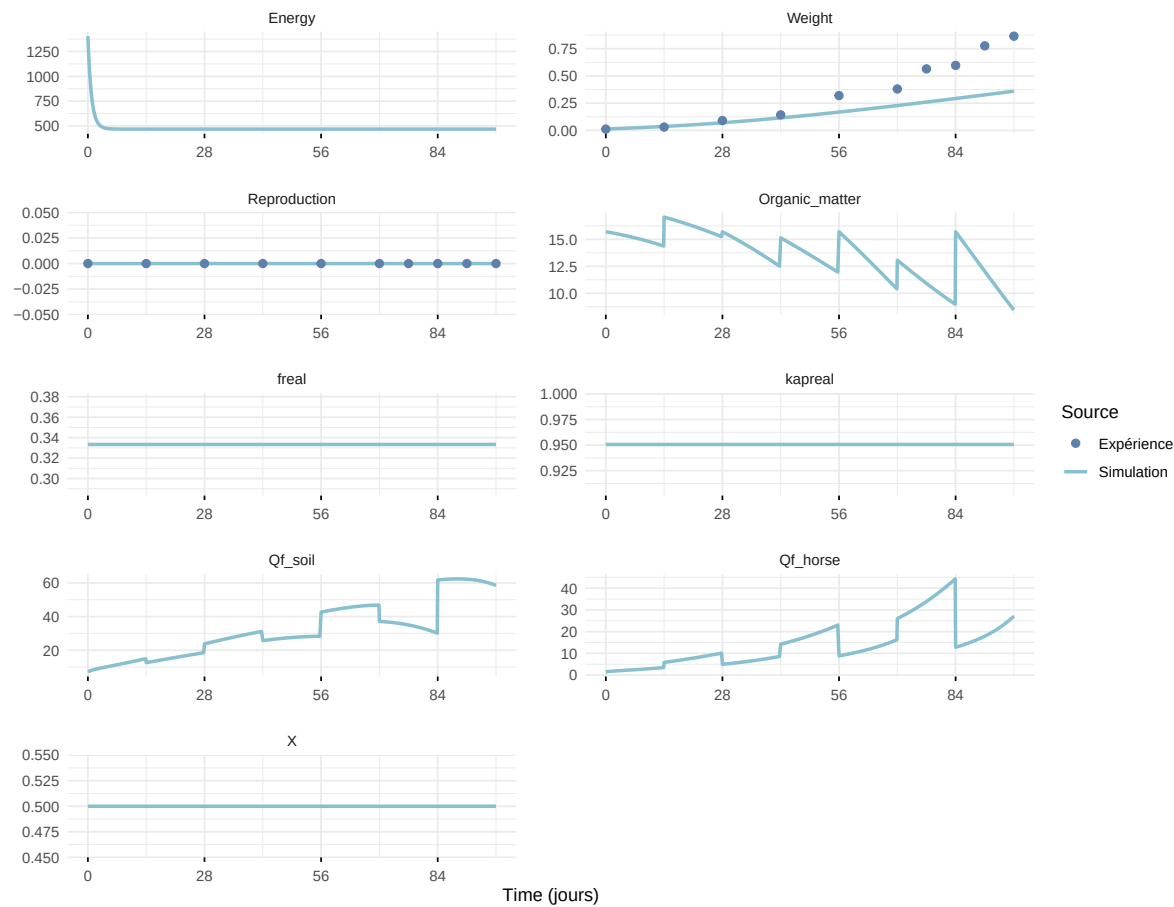
Expérience : Bart2019_XP4_2



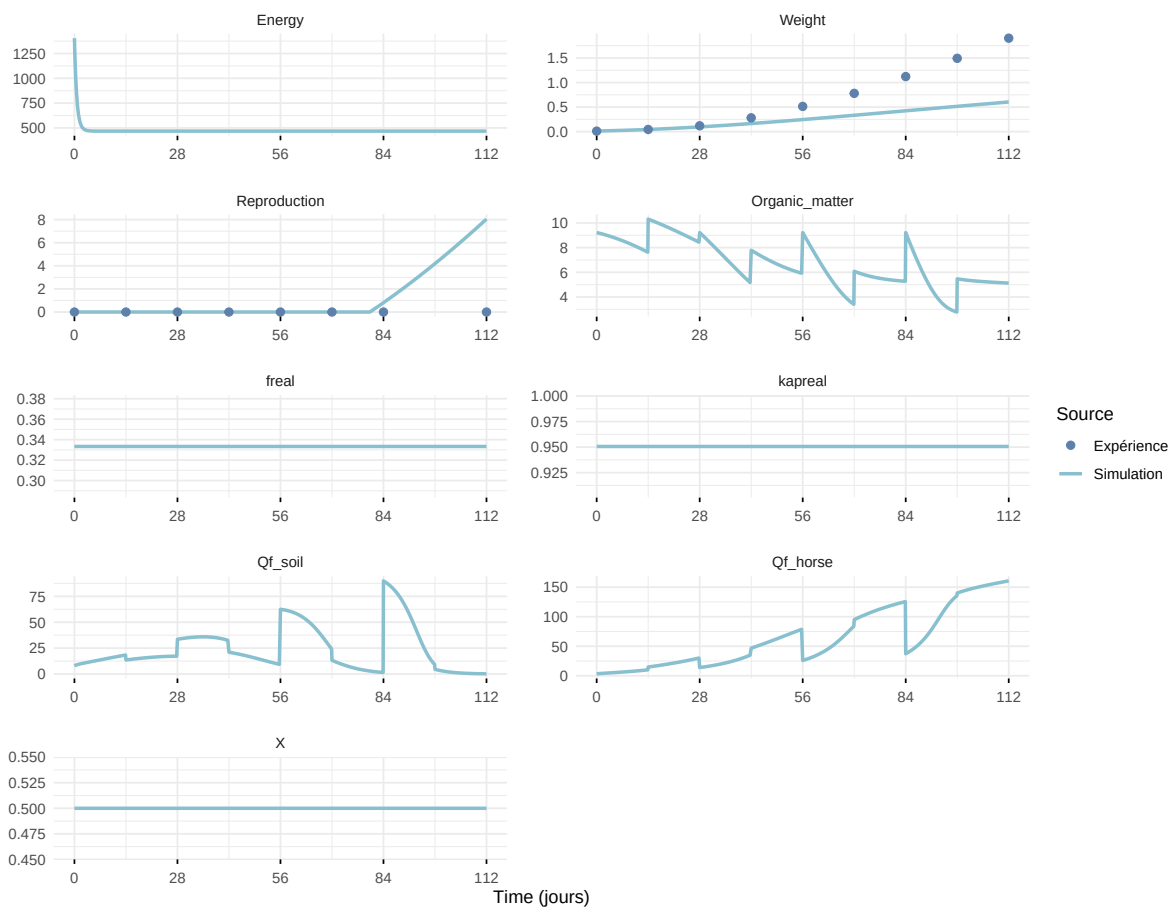
Expérience : Bart2019_XP5



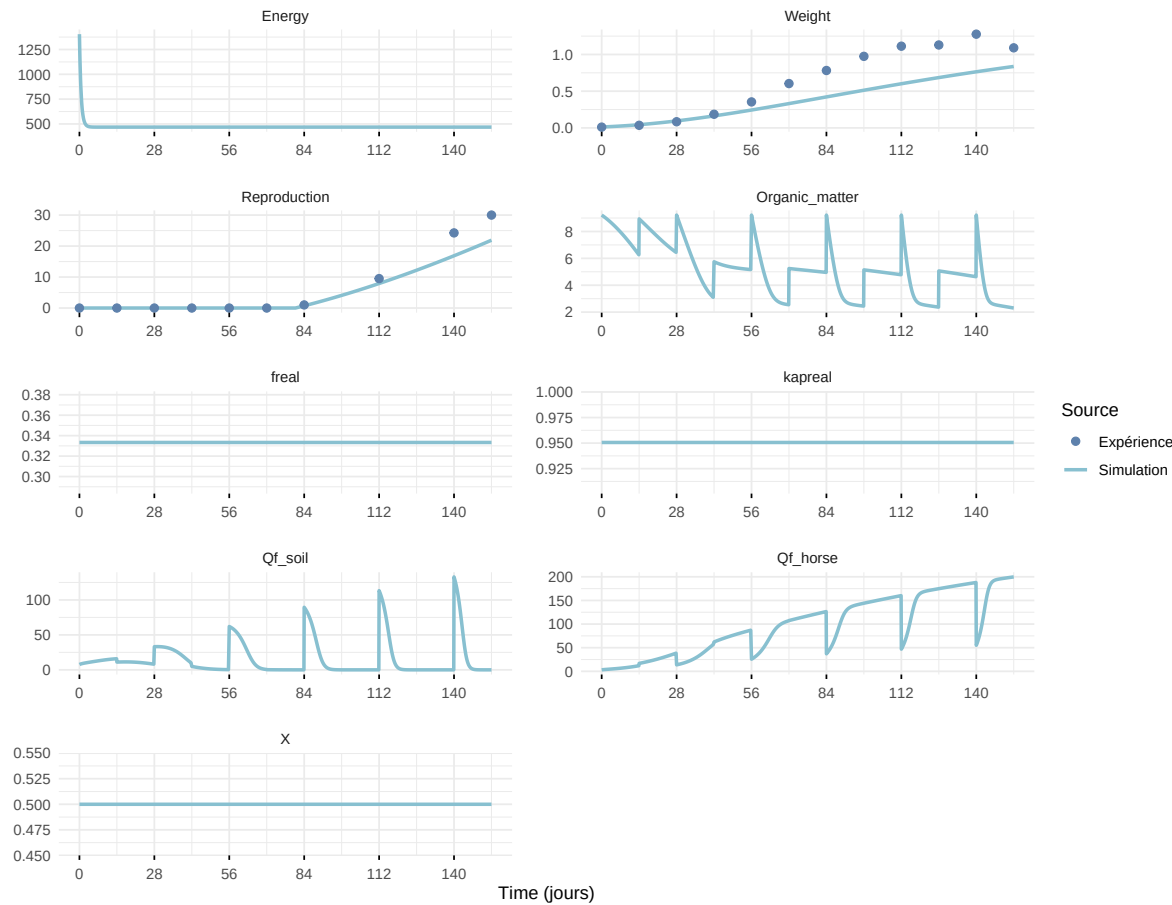
Expérience : Bart2020



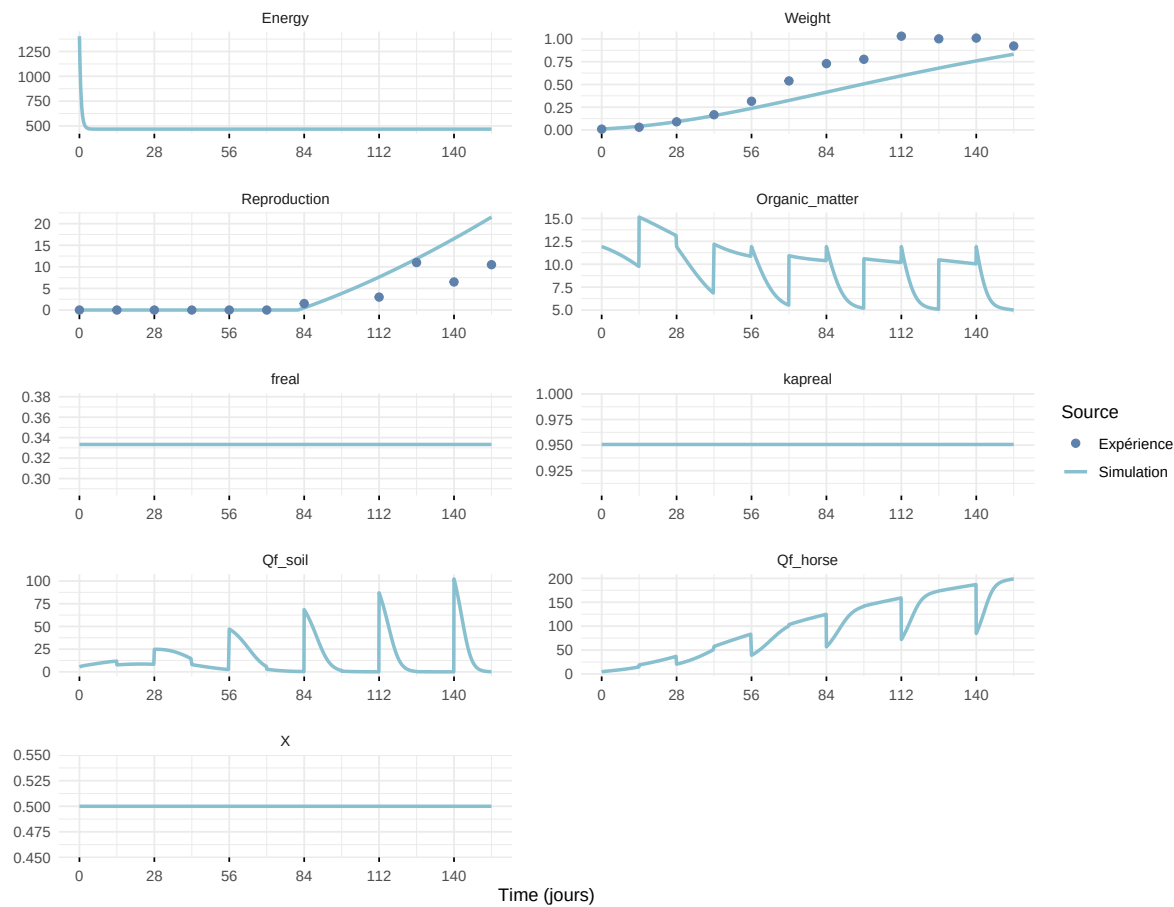
Expérience : Gollot2026_dens_D1



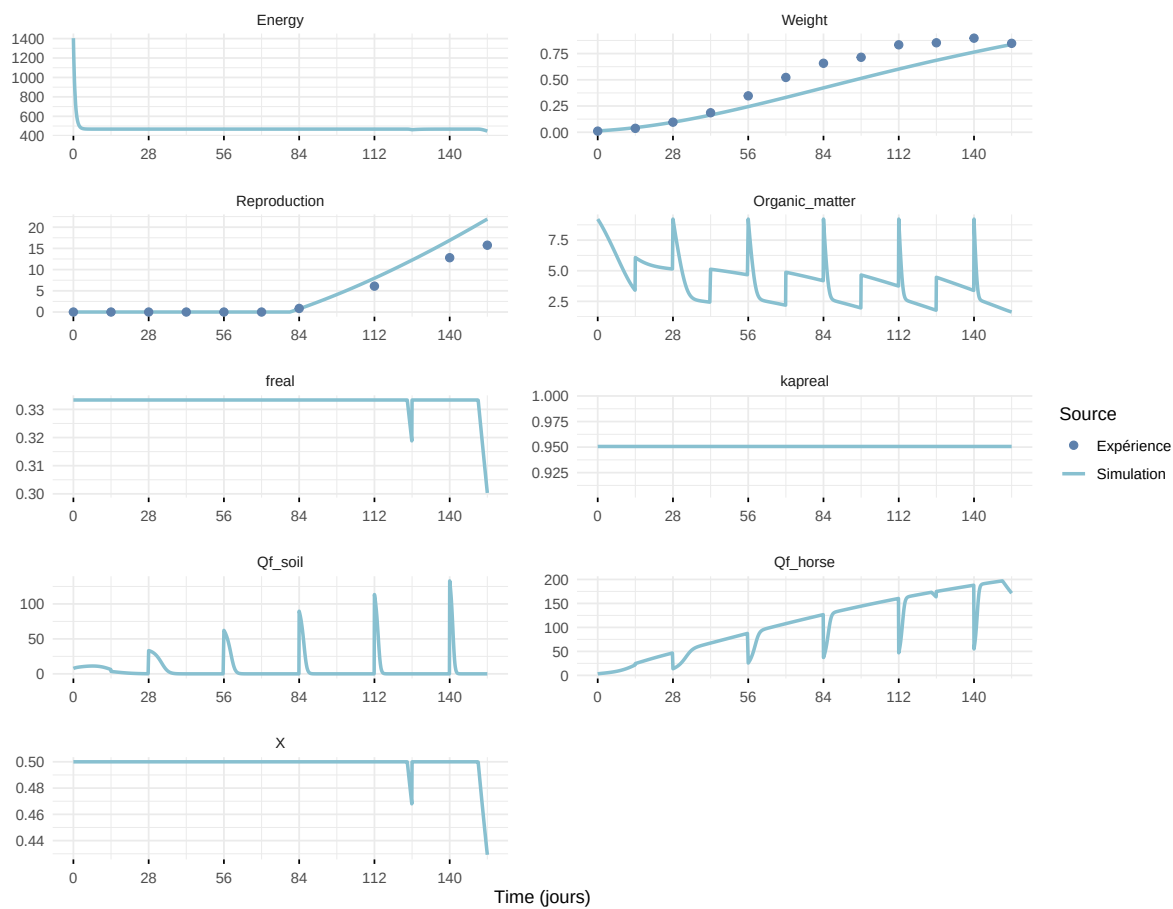
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

